CNN-LSTM daily update

13 October 2026 | 42-step time span | 10 training epochs

PROGRESS

Completed the evaluation pipeline using JPX-style Sharpe and Rank IC alongside MAE. The 10-epoch run produced nonconstant predictions and was evaluated on the same training data.

RESULTS / IN-SAMPLE

Metric	Value	Metric	Value
Training avg. loss	0.015604	Evaluation avg. loss	0.015570
Sharpe	0.100575	Rank IC	0.004274
Model MAE	0.015570	Zero-baseline MAE	0.015564
MAE improvement	-0.038636%	Prediction-target corr.	0.052963
Prediction mean	0.000770	Target mean	0.000397
Prediction std.	0.001132	Target std.	0.023428

Values rounded to six decimal places; metrics supplied by the researcher.

Interpretation: The pipeline runs end to end, but MAE is slightly worse than the zero baseline and Rank IC is near zero. These in-sample results do not yet establish performance on unseen data.

NEXT STEPS

1. **Validation:** Use an expanding window and add early stopping.
2. **Efficiency:** Save model checkpoints for pipeline checks without retraining.
3. **Training experiments:** Track average loss over 20 time frames and explore a $(\text{total_days} / 20)^2$ training schedule; the design still needs refinement.
4. **Architecture:** Test the ensemble approach in the linked paper [1].

[1] Applied Sciences (2023), 13(7), 4644 - ensemble-model reference.